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Fuzzy Type-1 Triangular Membership Function Approximation Using Fuzzy C-Means

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Abstract*— *Fuzzy logic is a way of many-valued computing logic that deals with the truth values of the variables between 0 and 1, unlike the conventional Boolean logic. Membership functions are used to depict the fuzzy values of given variable. Though membership functions are determined through expert's opinion, however, the one estimated through heuristic algorithms is the preferable methods. Membership functions determined through statistical and knowledge engineering methods are usually application dependent and cannot be applied on different datasets. This research focuses on generating the parametric values of the triangular membership function using a novel method. Initially, the Fuzzy C-means algorithm is utilized to generate the parameters values of the Gaussian membership function. Using a set of equations, these values then estimate the parameters of the triangular membership function. The proposed method is applied to the quality of web services data. From the results it is verified that the new approach of generating triangular membership functions can be adopted.

Keywords*— *Fuzzy Logics, Fuzzy-C Means, Gaussian MF, Membership Functions, Triangular MF.

I. INTRODUCTION

A set having crisp boundary is called a classical set i.e. 0 or 1 or in other words true or false. As an example, a classical set could be a set of numbers greater than 5.

$$A = \{x \mid x > 5\} \tag{1.1}$$

Here it's a transparent boundary, x will belong to set A if it is greater than 5 and cannot be an element of set A if it is less than or equal to 5. Although classical sets are successful in many applications and within the field of computer science and arithmetic, they have failed to represent the nature of human thoughts and concepts which tends to be imprecise and abstract. According to equation 1.1, if we take A =" Tall person" and x =" height", than someone having a height of 5.001 ft is tall whereas someone having height of 4.999 ft does not belong to the category of tall people. This bivariant existence of classical set cannot reflect human definition of tall person. The problem occurs as consequences of sharp transition of crisp set between inclusion and exclusion in the set. Whereas, fuzzy set deals with the "degree of truth" and does not have any crisp boundary. Here the transition between inclusion (belong to set) and exclusion (not belong to set) is unspectacular; this steady transition is represented by membership functions which provide flexibility to fuzzy sets to model frequently used linguistic terms, like "temperature is low" or "water is cold" [1]. As Lofti Zadeh stated in one of his article entitled "Fuzzy Sets" [2] in 1965, "such imprecisely defined sets or classes play an important role in human thinking, particularly in the domains of pattern recognition, communication of information, and abstraction." [1]

Fuzzy logic could be a way that reflects human thinking and decision making. The fuzzy logic emulates human's decision making and reasoning approaches that features all possibilities in between true(1) or false(0). Lofti Zadeh observed that human decision making includes a range of possibilities between true and false that is very unlikely to computers. Such as:

Certainly True	Possibly True	Cannot Say	Possibly False	Certainly False

Fuzzy logic works on degree of truth/ level of possibility of input to achieve the desired output. Fuzzy sets theory was introduced by Lofti Zadeh in 1965, that became the basis foundation of fuzzy logic systems (FLSs). FLS offers techniques to illustrate and compute data and information that are not precise and uncertain.. Keeping in mind their ability to model linguistic and system uncertainties, FLSs have been implemented successfully in various real-world applications.[3]

Fuzzy logic is not only a logic that is fuzzy, but it is the logic that is used to explain fuzziness. Zadeh in 1973 said, "the closer one looks at a real-world problem, the fuzzier becomes its solution." This fuzziness is best represented by its membership function in the fuzzy inference system (FIS). In other words, the membership function depicts the degree of truth in fuzzy logic[4]. This shows the importance of membership function generation in FIS. FIS is a system that uses fuzzy set theory to map input into an output. Different ways of generating single membership function have been considered in [5] and [6]. Generating more than one membership functions can provide the user with multiple options to solve a certain problem[7]. Multiple membership functions have been generated using current based approach i.e. Complementary Metal Oxide Semiconductor (CMOS) [7]. In the best of author's knowledge, generation of more than one membership function using data or FCM has not yet been reported in the literature. In this research, an approach to generate gaussian and triangular membership function using FCM technique is proposed.

The remainder of this paper is structured as follows: Section 2 presents the background of the fuzzy set and review various existing MF generation approach using FCM. Section

3 presents the proposed methodology. Simulation and results discussion are presented in section 4. Finally, section 5 presents the conclusion and future direction.

II. Background of the Fuzzy Logic System

A. Fuzzy Sets and Systems

A Fuzzy set, as the name suggests, is a set with no crisp boundary, unlike to a classical set. That is to say, the convergence from “belonging to a group” to “not belonging to a group” is minimal, and this smooth and seamless transition is defined by membership functions that provide versatility in modelling of linguistic expression that are very widely used such as “temperature is low” or “water is cold”. Remember that fuzziness does not derive from the randomness of the respective set members, but from the unpredictable and imprecise nature of abstract thoughts and concepts. [7]

Lofti Zadeh initially proposed a concept of fuzzy sets (FSs), [8] that guided to the foundation of the fuzzy logic systems (FLSs). fuzzy logic system provides a framework for the representation and computation of unpredictable and imprecise data and information.[8] Bearing in mind their ability to describe and model linguistic and systematic uncertainties. fuzzy logic systems have widely been applied in various real-World applications. [9] The central feature of fuzzy logic systems, is fuzzy inference system (FIS). It comprises of four important parts: (a) fuzzifier, (b) knowledge base, (c) inference engine, and (d) defuzzifier. Structure diagram of fuzzy inference system is presented in figure

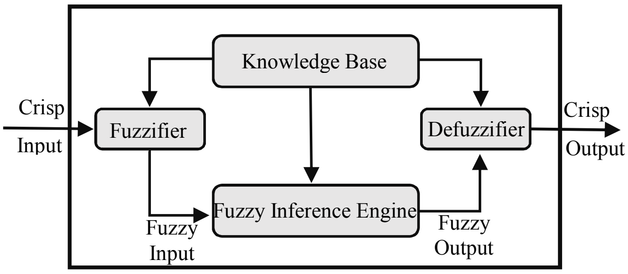

Figure 1: Fuzzy Inference System (FIS)

below.

In fuzzification stage, by look up into one or either several more membership functions, every data value is transformed to degrees of membership. [10] Fuzzy knowledge base comprises of fuzzy rules which all have a viable fuzzy relationship between input data and output data. If-Then conditions are used to demonstrate the rules in fuzzy knowledge base. [11] Fuzzy logic systems comprises of two rule base types which are: 1) Mamdani 2) Sugeno. [12] All the fuzzy inputs are converted into fuzzy outputs based on the fuzzy rules set at fuzzy inference engine. [12] Finally, at defuzzification stage all the fuzzy outputs are transformed into numbers and displayed in form of crisp output. [14]

Fuzzy inference system uses fuzzy logic and fuzzy set to represent the knowledge in interpretable form. fuzzy inference system is an attractive way used in many problems in many felids, like control, decision making, process models, and pattern classifiers [9]. Fuzzy Inference System has two parts, the first part converts the number of values (input/output) to an antecedent, consequent of IF…THEN rule. Each value can be divided to some ranges (the fuzzy set) with using certain labels to name these ranges. This part of a fuzzy inference system must choose one of the available kinds of membership functions, in addition to choosing a fuzzy connective tool like (And, Or) which determine their function. The second part begins by using the tools chosen before. By adopting one of the fuzzy inference systems model the system output can gain. This output is a result of the aggregation of the rules found in the knowledge base [10]. Since this paper focuses on fuzzy membership function, therefore, the other components of fuzzy inference systems will not be discussed.

B. Types of Fuzzy Membership Functions

A slope that describes how each point between 0 and 1 is mapped to membership degree is known as membership function (MF). Membership Functions are used to graphically illustrate the situation. Fuzzy logic systems deal with membership degree and degree of truth. Fuzzy membership functions can be described in several different ways, but they must follow the rules of fuzzy set description. Fuzzy set is determined by the shape and nature of membership function used, selection of fuzzy shaped is done based on the intent of use. [18] Fuzzy logic uses several different forms of membership functions i.e. triangular, trapezoidal, sigmoid and gaussian etc. Choosing the membership function depends on having fewer parameters or depends on the dataset being used. Straight lines are used to form simplest membership functions. Triangular and Trapezoidal membership functions are extensively used particularly in real time applications because of their simple mathematical formulas and efficiency of computation. [9] Nonetheless because the membership functions consist of straight lines, they are not seamless and smooth at the parameter-specified corner points. [19]

1. Triangular Membership Function

Three parameters {a, b, c} defines a triangular membership function.

$$\mu_F(x;a,b,c) = \begin{cases} 0; x \leq a \\ \dfrac{x-a}{b-a}; a < x \leq b \\ \dfrac{c-x}{c-b}; b < x < c \\ 0; x \geq c \end{cases} \quad (2.1)$$

An alternate expression can be presented for the previous equation by using min and max:

$$\mu_F(x;a,b,c) = max\left(min\left(\frac{x-a}{b-a},\frac{c-x}{c-b}\right),0\right) \quad (2.2)$$

The parameters {a, b, c} with (a < b < c) specifies the x-axis coordinates of triangular membership function.

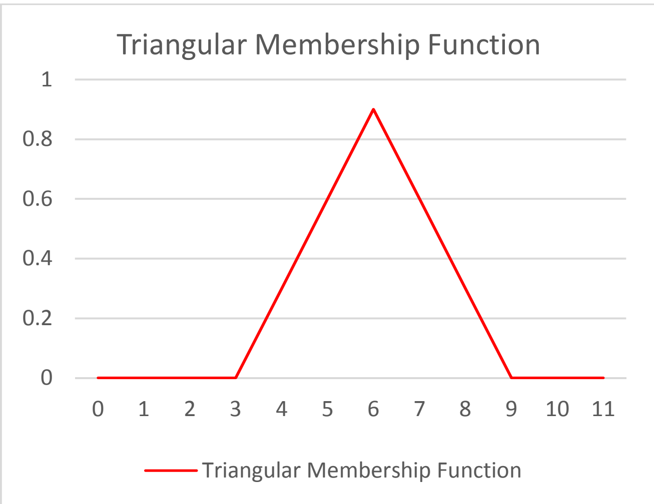

Figure 2: Triangular Membership Function

2. Trapezoidal Membership Function

Four parameters {a, b, c, d} defines a triangular membership function.

$$\mu_F(x;a,b,c,d) = \begin{cases} 0; x \leq a \\ \dfrac{x-a}{b-a}; a < x < b \\ 1; b \leq x \leq c \\ \dfrac{d-x}{d-c}; c < x < d \\ 0; x \geq d \end{cases} \quad (2.3)$$

An alternate expression can be presented for the previous equation by using min and max:

$$\mu_F(x;a,b,c,d) = max\left(min\left(\frac{x-a}{b-a}, 1, \frac{d-x}{d-c}\right), 0\right) \quad (2.4)$$

The parameters {a, b, c, d} with (a < a < c < d) specifies the x-axis coordinates of trapezoidal membership function.

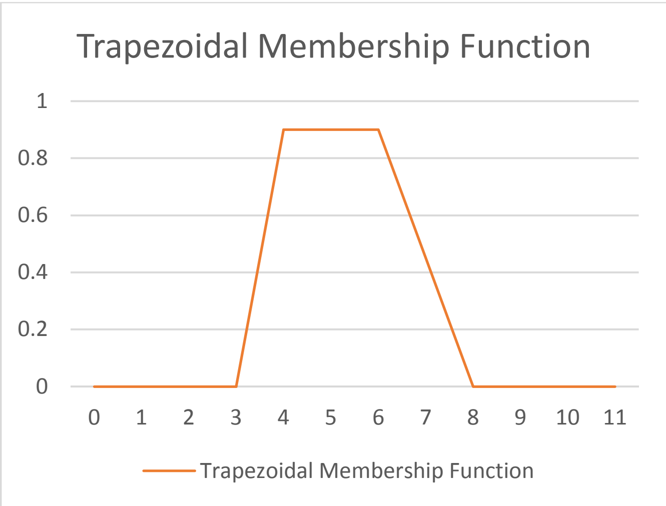

Figure 3: Trapezoidal Membership Function

3. Gaussian Membership Function

Two parameters (σ, c) defines gaussian membership function

$$f(x;\sigma,c) = e^{\frac{-(x-c)^2}{2\sigma^2}} \quad (2.5)$$

A gaussian membership function is entirely defined by cluster center c and width σ.

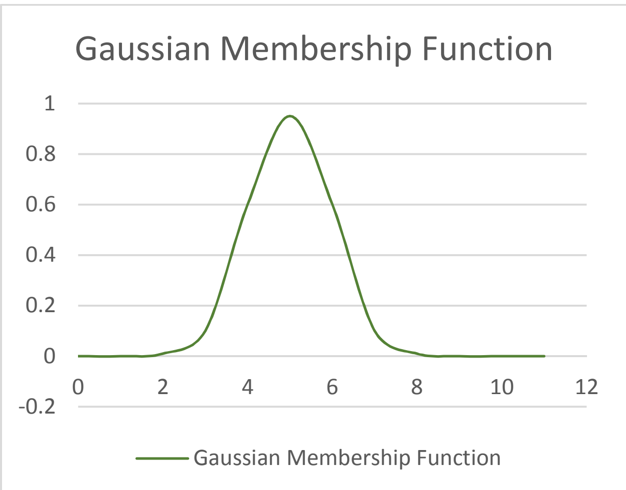

Figure 4: Gaussian Membership Function

There are two ways of generating fuzzy membership functions i.e. expert knowledge and generation through machine learning algorithms. Expert knowledge approach relays on expert in which system is analysed by expert and the parameters for construction of membership function is provided. The second approach uses machine learning technique are generates membership functions automatically. In this way of automatic membership function generation data processing techniques are being used like classification and clustering. The expert opinion approach possess some drawbacks such as lacking accuracy [20], incomplete opinion of expert [21] and unavailability of expert at all time. [22]. Moreover, it can be very subjective as different parameter values for membership function generation can be provided by different experts. [11]. Data processing method, on the other hand, can reduce, if not eliminate, those disadvantages of expert knowledge method. Hence, in the current study, we focus on the automatic generation of MF based on clustering.

C. Review of Fuzzy Membership Functions

Membership function represents the degree of truth in fuzzy logic [4], and this shows the importance of membership function generation in fuzzy logic system. Different ways of generating single membership function have been considered in [12] and [6]. Generating more than one membership function can provide the user with multiple options to solve a certain problem [13]. Multiple membership functions have been generated using Complementary Metal Oxide Semiconductor.(CMOS), in which electric current is passed as an input[13]. Most of the current ways of developing membership function using the Fuzzy C-means (FCM) clustering method will only produce single membership function that is gaussian membership function. This research will investigate how clustering can produce more than a single type of membership function that are triangular and gaussian membership functions. This is important as in some circumstances, one type of membership function might be better than another type depending on the problem and data. By having this new approach, we can have more options to choose the type of membership function for fuzzy inference systems.

By analysing Table 1, it can be observed that some problems can be more effectively solved by one type of membership function while others can have more accuracy if any other type of membership function is used. By generating more than one type of membership function, the user can have better options to choose the more suitable membership function and to solve the problem more effectively.

TABLE 1
EFFECTIVE FUZZY MF IN DIFFERENT CASE STUDIES

Problem	Effective MFs
Antenna Azimuth Position control. [14]	Triangular MF
Improving Decision Making in Crime Prevention[15]	Triangular MF
Nano-silica-concrete compressive-strength prediction[16]	Triangular MF
Non-stationary systems[17]	Gaussian MF
Mamdani based FIS for Decision Making[18]	Triangular MF Trapezoidal MF Gaussian MF

III. Proposed Methodology

The proposed approach of generating triangular membership function is depicted in Fig. 5 and is discussed below.

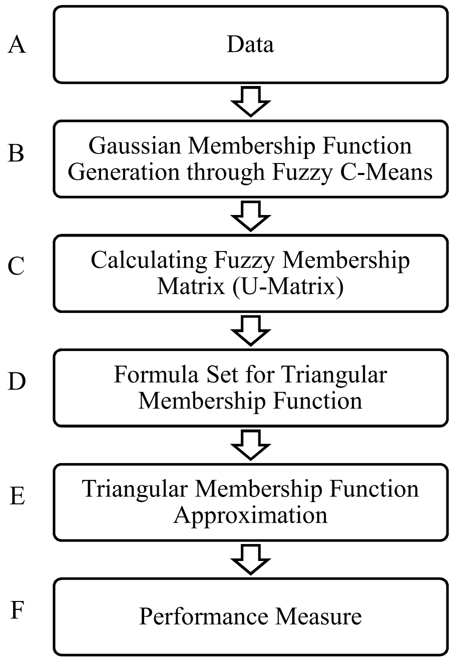

Figure 5: Flow of Proposed Approached

A. Data

The dataset utilized in this study is acquired from Mahmoud and Al Masri [19], [20].The data is gathered the available Web resources that include service-portals, sharing-platforms, service-registries, and search-engine. This was achieved by Web Service Crawler Engine (WSCE), that constantly crawls the internet to extract relevant QoS web services' data. Dataset used is comprised of actual real-time web services data obtained from the network. Dataset contain 3 variable attributes that are downlink time, uplink time and latency time for web services. Over 6000 real-time data point values are collected for downlink, uplink and latency time attributes. The data points are of millisecond (ms) in unit. Further, 3 synthetic data sets are created for all 3 attributes by adding and subtracting random values between 0 and 3 for constructing significance difference testing datasets. Each attribute is considered as an input for membership function generation. Thus, this research has 3 input for which three membership functions will be considered.

B. Gaussian Membership Function Generation through FCM

The approach for generating Gaussian is as the following:

- In the first step, data is input to the Fuzzy C-mean.
- Cluster centres are randomly selected by trial and error.
- New cluster centres are estimated through equation 3.1 and are stored in the equation matrix.
- Repeat the calculations until the threshold value is reached, which is the minimum targeted value.
- The initial randomly selected values are replaced with this calculated sets of the membership function (cluster centres).
- After that, the same process is repeated for calculating membership degree.
- Consequently, the final cluster sets, and membership degrees generated by Fuzzy C-means ultimately reflect the optimal clustering status of the underlying set of data.
- Fuzzy C-mean's outputs, namely cluster centre c, and membership degree matrix, U are used to create monitoring model's input membership functions.
- Fuzzy gaussian sets in input membership functions is defined by the equation below.

$$f(x;\sigma,c) = e^{\frac{-(x-c)^2}{2\sigma^2}} \qquad (3.1)$$

It can be observed from the equation above that fuzzy gaussian sets are defined by the centre of cluster c and the width property σ. Figure below illustrates fuzzy gaussian structure where centre and width are represented by c and σ. respectively.

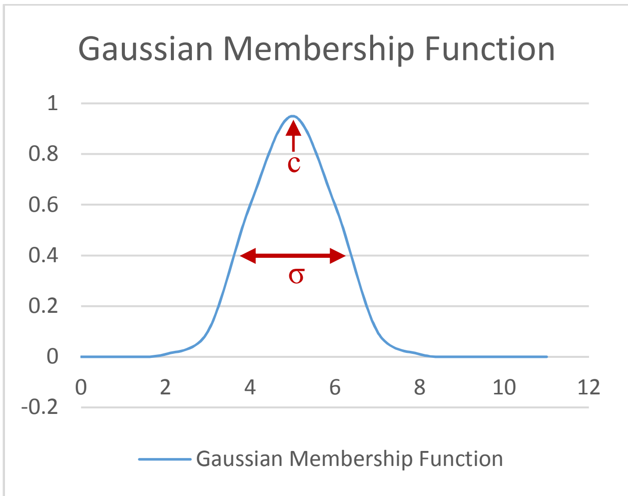

Figure 6: FCM Gaussian Curve

However, σ values are calculated on the basis of FCM membership degree outputs and cluster centre c. The values of data point, x also decides certain values.

$$\sigma = \frac{\sum_{n=1}^{k}\sqrt{(-(x_n - c_i)^2/(2 * \log(U_n))}}{k} \qquad (3.2)$$

Here k represents number of data points.

Depending on σ values of the datasets, and the cluster centres, the final fuzzy clustering outputs real-world.

C. Fuzzy Membership Matrix (U-MAtrix)

Fuzzy membership matrix (U-Matrix) also known as partition matrix contains the values of membership degree that is calculated based on the distance of each data points from the cluster centre. Dataset is passed through fuzzy C-means algorithm and its U-matrix is calculated as well as its cluster centres. U-matrix in FCM represents how data points in input space are mapped between 0 and 1. Equation 3.3 show the representation of fuzzy membership matrix.

$$Uik = \sum_{j+1}^{c} [(\frac{d_{ik}}{d_{jk}})^{\frac{2}{m-1}}]^{-1} \quad (3.3)$$

D. Formula Set

Once we get the U-Matrix and cluster centres, it is passed through a formula set to approximate the parametric values for triangular membership functions. The formula set being used in this approach is given below:

$$a = \alpha - \beta * \gamma;$$
$$b = \alpha; \quad (3.4)$$
$$c = \alpha + \beta * \gamma;$$

Here, a and c represent the floor value of triangular membership function while b represent the peak value of triangular membership function. α and γ are the values calculated from U-matrix and cluster centres where as β is constant.

E. Triangular Membership Function Approximation

Initially, a fuzzy set with triangular membership function is defined. Three parameters a, b and c specify a fuzzy set F. The degree of membership of pattern x to triangular fuzzy set F is given by:

$$\mu_F(x) = \begin{cases} 0; x \leq a \\ \dfrac{x-a}{b-a}; a < x \leq b \\ \dfrac{c-x}{c-b}; b < x < c \\ 0; x \geq c \end{cases} \quad (3.5)$$

or, more compactly:

$$\mu_F(x) = max\left(min\left(\frac{x-a}{b-a}, \frac{c-x}{c-b}\right), 0\right) \quad (3.6)$$

Three vertices of triangular membership function are defined by a < b < c. Membership values are computed for each value of x. Different formulas will be applied of the parameters for generation of membership function, Trial and error approach will be adopted and the membership function under consideration will be checked for accuracy against different datasets and results will be simulated to validate their accuracy.

F. Performance Measure

Triangular membership function is generated and is FIS is created based on fuzzy rules set and significance difference test is performed in order to test how accurately it handles the uncertainty. Results of the testing are explained in next section.

IV. Simulation and Results Discussion

A. Simulations

The estimation of gaussian and triangular membership The estimation of gaussian and triangular membership functions based on the proposed approach can be seen in Figure 7 and 8, respectively. The plot in Figure 7 below shows three gaussian membership functions generated based on FCM clustering. Y-axis represents fuzziness between 0 and 1, while x-axis depicts membership degree. Here, red colour plot shows a cluster with low value whereas blue and yellow plots show medium and high clusters, respectively.

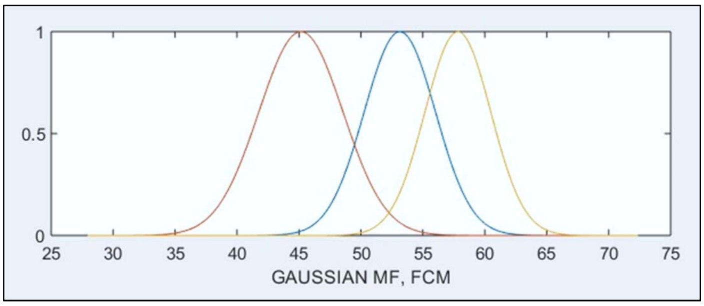

Figure 7: Gaussian MF Estimation

After the successful generation of Gaussian membership function, the clustered data is converted into a Triangular membership function using the proposed approach. The final FCM based triangular membership functions are presented in figure 8.

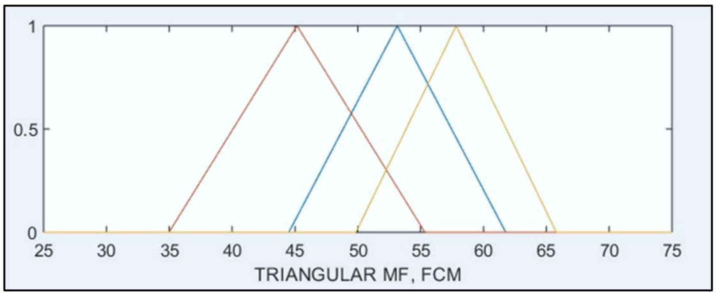

Figure 8: Triangular MF Estimation

FIS is developed based on Gaussian and Triangular membership functions and tested against test dataset, dataset used for testing is designed based of the distribution of membership points. In case of the MF generated in figure 8, 100 equally distant test data points are calculated between the range of 42 and 64 that shows the densest points distributed area. To check the accuracy of handling uncertainty, significance difference test is done on 3 synthetic datasets to validate the results. The results of the test show that the significance difference of Gaussian and Triangular is approximated to each other that shows the new way of generating Triangular membership function can be adopted for membership function generation. The result of the significant difference test is shown in table 5.

TABLE 2
SIGNIFICANT DIFFERENCE TEST

FIS Name	Synthetic Data 1	Synthetic Data 2	Synthetic Data 3	Total Difference	Percentage %
Triangular	381.0029	311.3397	157.0561	849.3983	**0.168666**
Gaussian	419.1609	308.0986	123.1637	850.4233	**0.168869**

It can also be observed from table 4 that for synthetic data 1, Triangular membership function shows better results whereas Gaussian membership function is more significant for synthetic data 2 and 3. This proves the importance of having more than 1 membership function as the selection of membership function is also dependent on data set. Result of t-test is presented below to find significant difference.

TABLE 3
t-TEST FOR SIGNIFICANT DIFFERENCE

t-Test: Paired Two Sample for Means for Triangular and Gaussian Membership Function		
	Triangular MF	*Gaussian MF*
Mean	283.1327775	283.4744607
Variance	13134.69952	22358.34301
Observations	3	3
Pearson Correlation	0.997431511	
Hypothesized Mean Difference	0	
df	2	
t Stat	-0.016367026	
P(T<=t) one-tail	0.49421377	
t Critical one-tail	2.91998558	
P(T<=t) two-tail	0.98842754	
t Critical two-tail	4.30265273	

As the p value is greater than 0.05 which shows that the approximation results of estimated triangular membership function are not significantly different.

V. Conclusion

In this research, an approach to generate Gaussian and Triangular membership functions using Fuzzy C-means clustering is proposed that will enable the user to have multi solution set for solving data-science and forecasting problems. The approach will enhance the accuracy of the results. This research is focused to identify the parameter values and propose an approach to generate Gaussian and Triangular membership functions using Fuzzy C-means clustering algorithm. The FIS generated using the proposed approach will be applied to data-science and forecasting problems to validate the effectiveness and accuracy of the results of respective membership functions.

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